

UAV Navigation

World models · vision–language agents · 3D Gaussian scene memory

A short survey of where the field stands: the mechanisms that exist, what they actually establish, and the architecture that is still missing for aerial autonomy.

Autonomy is moving toward predictive agents

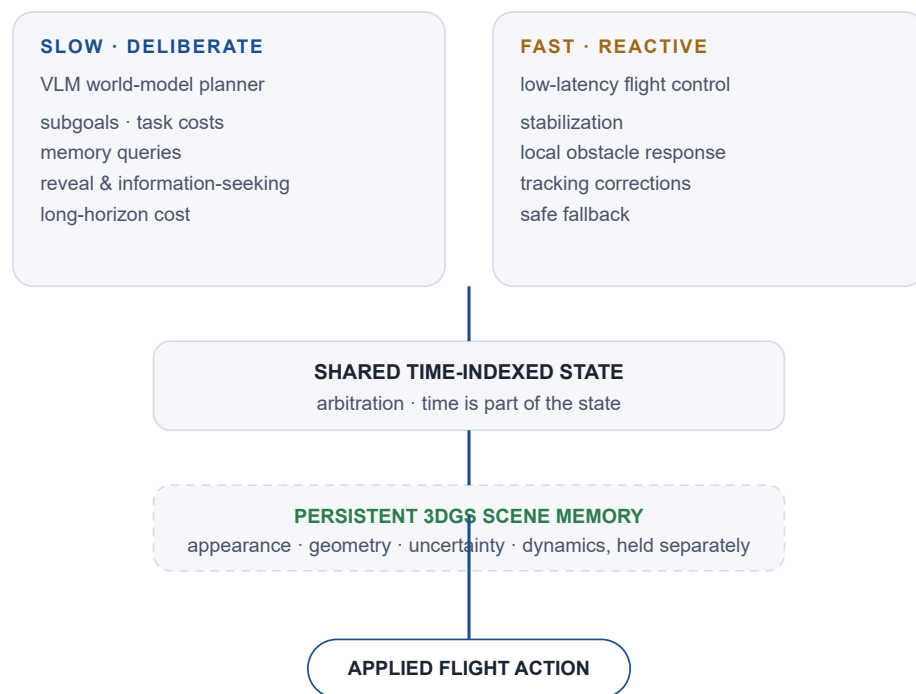
The field is not converging on a single architecture. Capabilities are maturing in parallel, but the systems that actually fly today rest on classical geometric autonomy; the learned predictive stack reviewed here has not yet been deployed at scale.

CAPABILITY	WHERE IT STANDS
Vision–language navigation	models interpret instructions and select actions, but performance drops sharply on unseen scenes and long-horizon tasks
World models	latent and generative models support planning from imagined futures, but prediction is not control sufficiency
Aerial world–action models	predict future geometry, views, or trajectories together with actions; language grounding remains a separate problem
3D scene memory	Gaussian Splatting provides view synthesis, semantic mapping, and simulation, but a visually convincing map is not collision geometry
Asynchronous autonomy	fast/slow branches and action-chunk correction target inference delay, yet action age is still rarely measured
Safety & uncertainty	barrier functions, certified maps, and calibrated action sets exist but are rarely integrated with learned planning

Each capability has advanced largely in isolation; the central research problem is to connect them into one working system.

The central research problem

The goal is not simply to combine a **VLM**, a **JEPA**, and a **drone**. It is to learn a **predictive state** that preserves what the task depends on: geometry, ego-motion, visibility, identity, the consequences of actions, uncertainty, and the language-conditioned goal, while the aircraft keeps moving and observations arrive late.



A practical frame is a **two-timescale agent**:

- A fast reactive controller handles **immediate flight**.
- A slower, VLM-configured planner thinks **ahead** by choosing subgoals, seeking information, and estimating long-horizon cost; it supervises the fast branch or distills policy into it.
- A persistent scene memory supports both branches, keeping appearance, geometry, uncertainty, and dynamics as **separate quantities**.

Often the best action reveals information rather than simply reducing the distance to the goal. This is a central feature of aerial search.

Tracking, navigation, and exploration set different deadlines

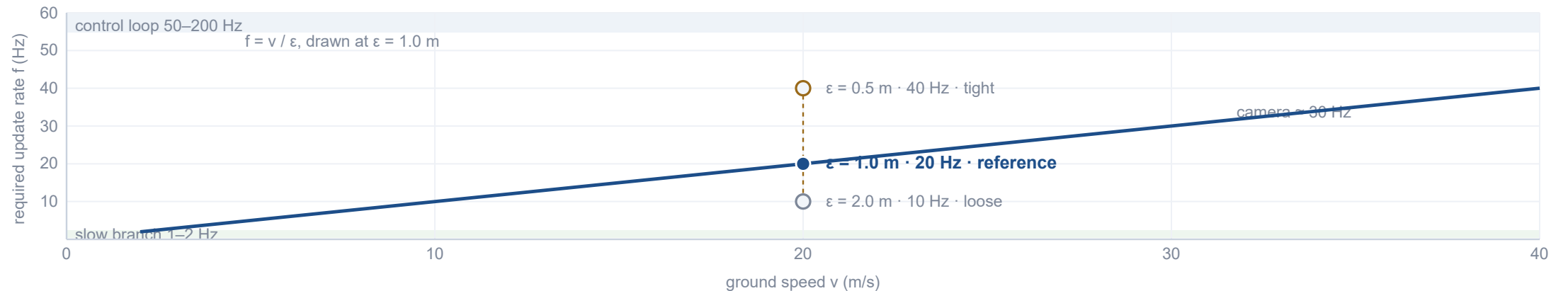
These are some motivating examples of missions a UAV might be asked to carry out.

MISSION	WHAT SUCCESS MEANS	WHAT MUST BE FAST
Navigation	reach the correct destination under instruction, time, and safety constraints	get the decisive evidence before the route choice becomes costly to reverse
Tracking	keep following the correct moving target, including through temporary loss of view	motion and perception timing that preserve a reachable recovery view; belief that updates before the target slips away
Exploration	discover trustworthy space within a resource budget and keep a feasible return route	belief updated on independent evidence before an inferred connection is trusted

Latency matters in every mission, but the deadline differs: route reversibility for navigation, target recoverability for tracking, and evidence independence for exploration.

How fast must the models run?

How often the model must update follows from $f = v / \epsilon$ (in Hz, for ground speed in m/s and a motion budget ϵ in metres): speed sets the x-axis, and ϵ decides how often you must update. The line below is drawn at $\epsilon = 1.0$ m.



The third variable ϵ sets how often you must update at a given speed: track tightly ($\epsilon = 0.5$ m) and you double the reference rate; cruise loosely ($\epsilon = 2.0$ m) and you can halve it.

Rate becomes comparable only when paired with speed. The distance per update, $\epsilon = v / f$, is what the number means for the vehicle; report it, not Hz alone.

What each component learns, and what it does not establish

Different works study different components, and a model can be built for one job without establishing what a UAV actually needs.

OBJECT	IT LEARNS	IT DOES NOT ESTABLISH BY ITSELF
Visual representation	features of images, video, geometry	controllable dynamics, memory, metric flight feasibility
JEPA / latent predictor	future representations, conditioned on context & actions	correct counterfactuals outside training coverage
Generative world model	future images, video, scene observations	low latency, calibrated uncertainty, safe control
VLM	relations between visual evidence and language	action consequences, metric accuracy, flight stability
VLA	actions conditioned on vision–language–history	an explicit model for replanning or diagnosis
World–action model	coupled future prediction and action generation	general language understanding, cross-mission transfer
3DGS map	view-dependent appearance, spatial primitives	complete geometry, unknown-space safety, dynamics
Belief model	alternative hidden states and uncertainty	correct belief updates or useful information-gathering

A *name* a building without knowing how to approach it; a 3DGS render can treat unknown space as free; a VLA can imitate a route without being able to revise it after occlusion. None of this means the method is bad; each component serves a different purpose.

VLM

can

World models & JEPA-style prediction

These foundations set the bar: a UAV predictor must clearly beat a small encoder trained from scratch, and must also improve on a pretrained representation used directly.

Established baselines

DreamerV3 learns a compact environment model, *imagines* futures, and acts on those imagined outcomes.

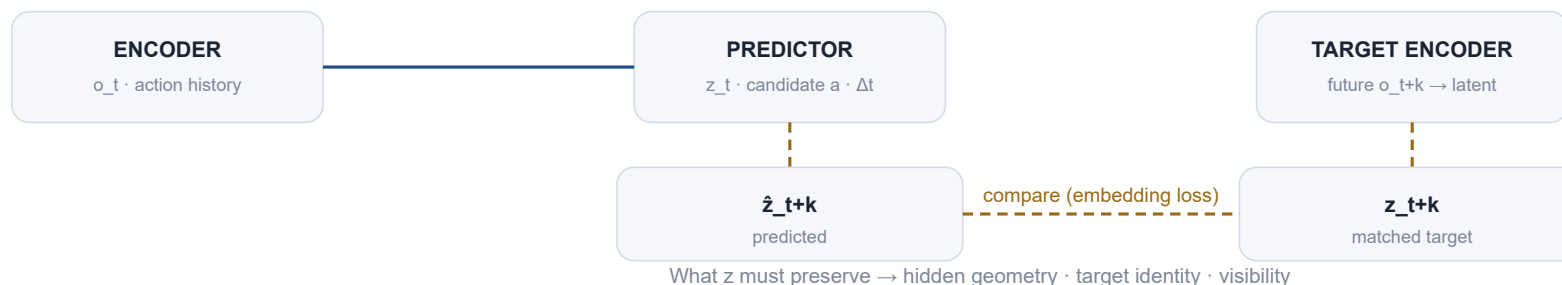
DINO-WM shows how a pretrained visual representation can support latent dynamics and goal-directed planning.

V-JEPA 2 makes the point that video-representation pretraining is *not* the same as action-conditioned control.

The decisive test

The research content is not the embedding loss; it is whether the representation supports the following:

- **action ranking** and replanning,
- task **transfer**, and
- **calibrated failure detection**.



What is already taken, and what stays open

GENERIC DIRECTIONS ALREADY COVERED

Factorized predictive state → Orthogonal JEPA, JEPA-Anything

“Add physical meaning” → PhyLatent, action-based representation learning

Object / masked prediction → Causal-JEPA

Multi-step consistency → Semigroup-JEPA, hierarchical latent WMs

Structured uncertainty → UWM-JEPA, belief-latent work

Counterfactual prediction → Twin Rollouts, controlled world-model theory

OPEN UAV QUESTIONS

- **Action sufficiency:** what must survive compression
- **Memory after compression:** the small obstacle still matters
- **Partial observability:** belief, not one best map
- **Language → dynamics binding:** goals meet physics

Average rollout error can hide the collapse of the rare future that actually decides the action.

Three architectures for vision–language navigation

The literature has converged on three patterns, which trade off control, safety, and debuggability differently; practice is moving toward hybrids of all three.

VLM AS PERCEPTION

The model produces captions, object locations, semantic maps, or labels, which a separate planner and controller then use for navigation.

VLM → semantic map → planner → controller

easy to deploy and inspect

semantic errors amplify downstream, and the model never sees the consequences of its actions

VLM AS HIGH-LEVEL PLANNER

The model proposes subgoals or route decisions, while MPC, sampling-based control, or a geometric controller handles the flight.

VLM → subgoals → MPC / geometry

clear safety interface

may select plans that are spatially or temporally infeasible unless feasibility is exposed to the model

END-TO-END VLA

A single model maps vision, language, and history directly to actions or action chunks; FlightGPT and AeroVLA exemplify this direction.

single network → actions

tight perception–action coupling

debugging, safety verification, and long-horizon replanning are harder, and so is transfer from simulation to real flight

The trend is hybrid: semantic slow branches, fast action branches, recurrent memory, waypoint interfaces, action chunks, and explicit uncertainty. This reflects onboard compute and delay constraints, not evidence that one architecture has won.

Capable pieces, but no unified agent

The field is fragmented: no current result demonstrates an agent that does all of the following at once.

CAPABILITY	REPRESENTATIVE WORK	WHERE IT STANDS
Aerial VLN benchmark	OpenFly	shared evaluation infrastructure; not a strong baseline or a real-flight guarantee
Language → action	FlightGPT · AeroVLA · Qwen-RobotNav	grounds language to waypoint/action output; long horizon, calibration, unseen scenes remain hard
Future + action	WorldFly · FlowPilot	predicts future visual/depth with trajectories; costly ; language and action sufficiency are separate
Aerial world model	SkyJEPA	action-conditioned quadrotor dynamics, sim-to-real direction; dynamics ≠ camera-based navigation
Fast / slow inference	FSD-VLN · LiteVLA-H · AsyncVLA	decouples reasoning from control; action age is rarely measured end-to-end
Memory for VLA	ReMem-VLA, recurrent-query	history improves temporal grounding, but may keep appearance, not belief or state
Open-vocab object nav	AirHunt · AeroBelief · AECNav · ConsistNav	semantic search, evidence consolidation, belief; verification & occlusion recovery under-tested

Within the work reviewed here, no single system yet combines language grounding, a calibrated spatial belief, action-conditioned prediction, occlusion-aware planning, flight dynamics, and transfer to new scenes.

Language plays three different roles

Treating language as one undifferentiated input produces agents that rewrite the physics when the instruction changes, or fail to update their belief when the evidence does.

01 · TASK SPECIFICATION

Selecting a destination, a target, or an inspection objective.

changing the instruction changes the **task cost or subgoal**, not the physics

02 · EVIDENCE

Reports such as “route blocked,” “target moved,” or “surface inspected.”

new evidence should **update both the belief and the future prediction**

03 · KNOWLEDGE OF DYNAMICS

Properties that change how objects and the environment behave.

carries constraints on **how things move**, conveyed in language

Most VLM/VLA systems do not expose enough structure to test whether an instruction change really stops at the task layer.

A Gaussian map is a memory, a simulator, or a state

3D Gaussian Splatting serves three roles of increasing strength; the strongest is also the least understood.

PERSISTENT MEMORY

It stores appearance, semantics, and spatial relations across viewpoints, which is useful for search and inspection from changing altitude and angle.

VISTA · ATLAS Navigator

SIMULATOR

It provides fast novel-view synthesis and a visually rich environment for training.

SOUS VIDE / FiGS · GRaD-Nav

PREDICTIVE STATE

Gaussian features feed a learned controller or world model; this is the strongest role and the least verified.

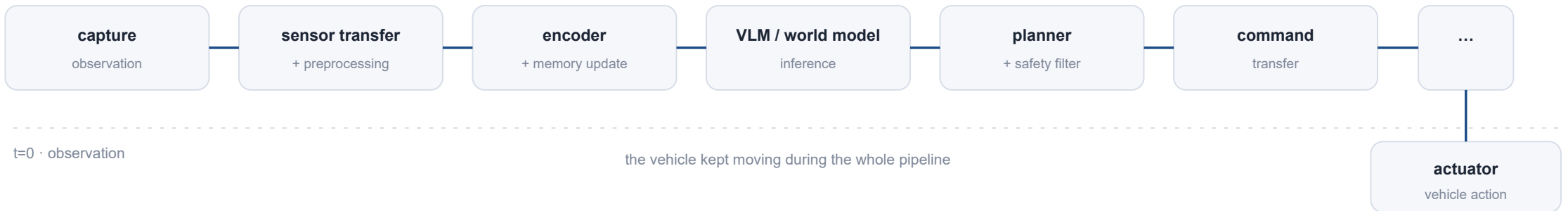
must keep collision geometry, ego-motion, visibility, consequences

A plausible render in unobserved space is not a safe occupancy estimate. The map therefore needs an explicit interface to geometric safety, state estimation, dynamic objects, and unknown space.

Time is part of the state

The aircraft keeps moving while the model reasons, so a plan computed from an old observation can be unsafe even when the model is exact at the time of its input.

END-TO-END ACTION AGE, NOT “MODEL INFERENCE TIME”



REPORT, NOT ASSUME

Neural inference time is not the control-loop latency: a model reporting sub-20 ms inference can still act on a much older observation. Reports should include timestamped action age, motion during inference, and dropped or superseded actions.

DEFENSIBLE SHAPE

A **fast reactive actor** (stabilization, local obstacles, tracking, fallback) plus a **slower deliberative branch** (subgoals, costs, memory queries, reveal maneuvers), sharing a time-indexed state with explicit arbitration.

Occlusion draws the line between memory and belief

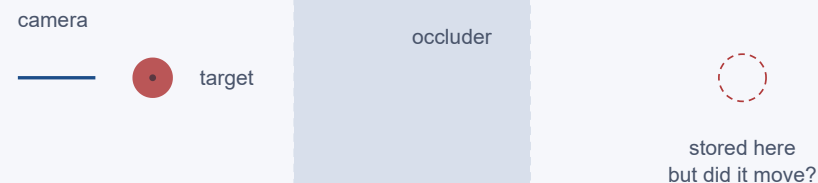
Remembering the last location of a target is not enough when the target may have moved. A capable agent must keep alternatives, predict how an action changes visibility, and update its belief on evidence.

- **Memory** stores the last-seen state. **Belief** holds a weighted set of alternatives.
- A useful model predicts two things: how the **scene evolves** under a candidate action, **and** what **evidence** that action exposes and how belief changes.
- This is stronger than an exploration bonus or a confidence head.

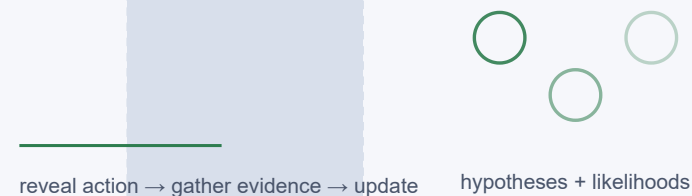
DESIGN RULE

Keep **imagined evidence** (produced inside the planning branch) separate from **observations admitted into live memory**.

MEMORY · LAST SEEN



BELIEF · COULD BE HERE



What the learned part proposes, what the safety layer enforces

Control Barrier Functions, backup trajectories, geofencing, certified mapping and perception-aware MPC give explicit constraints a language model must not be allowed to disable.

LEARNED · PROPOSES

task value · information gain · inspection value · subgoals
a VLM may configure the task-facing cost, never the safety terms

SAFETY · ENFORCES

control barrier functions · backup trajectory · geofence · vehicle limits
sensor validity · recoverability · collision and clearance constraints

AIRCRAFT + ACTUATORS

physics · limits · network · inner control loop

A language command cannot disable a barrier function.

Two errors recur: a valid rendered depth sample is treated as proof of free space, and a nominal controller period is treated as sensor-to-actuator latency. The assumptions must be stated, or the proof does not cover the real stack.

Seven concrete research directions

Each is a concrete mechanism that can be tested on route-level UAV tasks, rather than a vague ambition.

PREDICTION & MEMORY

- A.** Action-preserving, task-conditioned world models: a predictive state that preserves route feasibility, visibility, and safety-critical geometry when the task changes.
- B.** Belief dynamics for information-seeking flight: predict the future scene, the evidence an action reveals, and the belief update that evidence should cause.

LANGUAGE & ADAPTATION

- C.** Achievable language intentions: “inspect the far side” should imply viewpoint, visibility, and motion requirements, not just an endpoint label.
- D.** Selective adaptation: keep scene, vehicle-response, and sensor models separable, so that a change in payload, lighting, or wind updates one without corrupting the others.

SYSTEM & SAFETY

- E.** Continuous prediction under delay: a time-indexed workspace that reconciles late results and rejects or repairs stale plans.
- F.** 3DGS with control semantics: occupancy confidence, unknown space, and dynamic objects, comparing view quality, collision prediction, and route success separately.
- G.** Safety-aware learned planning: learned task value and information gain under formal constraints, with a recoverable fallback and explicitly stated assumptions and timing.

The novelty bar is high: “use factorized JEPA,” “add latent uncertainty,” and “use Gaussian tokens” are already represented. A contribution must justify the UAV-specific state, update rule, or control interface.

Where these systems are actually tested

The papers reviewed here run on a small set of platforms and datasets. Each gives different evidence, and the gaps explain why real-flight guarantees are rare.

ENVIRONMENT	WHAT IT PROVIDES	WHAT IT DOES NOT PROVIDE
Photoreal RGB simulation AirSim · Isaac Sim · Gazebo (PX4 / ArduPilot)	controllable scenes, ground truth, repeatable episodes	full network, actuator, lighting, and calibration realism; scenes are often frozen while the model plans
Gaussian renderers as simulators SOUS VIDE / FiGS · GRaD-Nav	fast novel views, visually rich training and evaluation	appearance is not collision geometry; metric scale and odometry transfer stay unproven
Benchmarks and datasets OpenFly aerial VLN	shared, large-scale route evaluation	offline success measures, not closed-loop flight or measured latency
Real aircraft PX4 / ArduPilot, onboard compute	true latency, power, and network behavior	expensive and rare in this literature; action age is seldom measured

Comparable results need a matched sensor suite, action space, scene split, and compute budget. Simulation success, inference speed, and rendering quality are not closed-loop real-flight autonomy.

Evaluate a UAV agent up a six-step ladder

Start from one real task, such as point-to-point navigation, target tracking, or occlusion-aware inspection, and evaluate upward through the following levels.

1	Perception & state	localization · geometry · target identity · map uncertainty
2	Prediction	one-step and multi-step action-conditioned latent prediction
3	Decision	action ranking · subgoal choice · reveal value · replanning
4	Closed loop	route success · final error · path length · collision / clearance · action age
5	Generalization	unseen scenes · weather / lighting · altitude · noise · moving objects · dynamics
6	Realism	motion during inference · timestamped commands · stale actions superseded · assumptions checked

MINIMUM BASELINES

a geometric controller or MPC, a native aerial VLA / world-action model, a predictive baseline in the style of FlowPilot or SkyJEPA, a fast/slow or asynchronous system, and a safety-filtered version.

ABLATIONS

remove language configuration, persistent memory, belief updates, future prediction, delay handling, and safety constraints, one at a time.

The question this literature sets up

Can a UAV hold a **persistent, uncertain scene model** and use a **language-configured predictive branch** to choose long-horizon actions, while a **fast reactive branch** keeps flight safe during delayed deliberation?

WHY IT IS REAL

Flying changes what the vehicle can see, and the best action may be one that reveals information rather than one that shortens the distance to the goal. The question therefore ties together VLM task interpretation, JEPA prediction, 3DGS memory, active perception, asynchronous control, and safety filtering.

MAKE IT DEFENSIBLE

A credible paper should claim **one** structural mechanism, such as action-preserving task-conditioned compression, belief-aware observation prediction, or continuous time-indexed inference, and demonstrate it on route-level tasks with measured delay and strong baselines.

Selected references

WORLD MODELS & JEPA

LeCun, *A Path Towards Autonomous Machine Intelligence*
DreamerV3 · DINO-WM · V-JEPA 2 · Dynalang
Orthogonal JEPA · JEPA-Anything · VL-JEPA · Gaussian-JEPA

AERIAL & VLA

OpenFly · SkyJEPA · FlowPilot · OpenVLA
FSD-VLN · AsyncVLA · LiteVLA-H · ReMem-VLA
AirHunt · AeroBelief · AECNav · ConsistNav

3DGS & MAPPING

VISTA · ATLAS Navigator · SemSafe-3DGS
FastBridge · Certifiably-Correct Mapping
semantic and uncertainty-aware Gaussian maps

SAFETY & CONTROL

Control Barrier Functions · backup trajectories
PA-MPPI · gatekeeper · certified mapping
perception-aware MPC · belief-space planning

Full citations, with venues and years, appear in the companion literature review document.